1.15 Graph Sketching: Cubic Graphs

Chapter 4 · Original slide 4 · Polynomial Graphs

In Chapter 2 we briefly saw that a polynomial expression is of the form:

$a + bx + cx^2 + dx^3 + ex^3 + \ldots$

where $a, b, c, d, e, \ldots$ are constants (which could be 0).

The order of a polynomial is its highest power.

Order	Name
0	
1	
2	
3	
4	
5	

These are covered in Chapter 5.

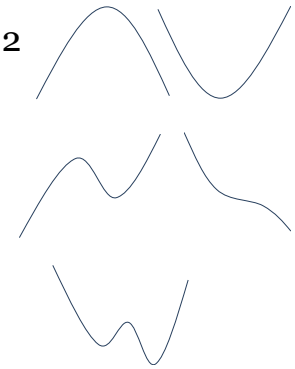
Chapter 2 explored the graphs for these.

We will cover these now.

While these are technically beyond the A Level syllabus, we will look at how to sketch polynomials in general.

Chapter 4 · Original slide 5 · Polynomial Graphs

2



What property connects the order of the polynomial and the shape?

Order:

Bro Note: ...Actually this is not strictly true, e.g. consider $y = x^4$, which has a U shape. But this is because multiple turns are being squashed into a single point.

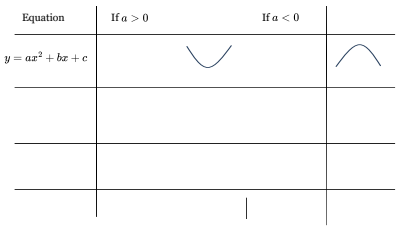
In Chapter 2 how did we tell what way up a quadratic is, and why does this work?

3

4

1.15 Graph Sketching: Cubic Graphs

Chapter 4 · Original slide 6 · Polynomial Graphs



e.g. If $y = 2x^2 + 3$, try a large positive value like $x = 1000$. We can see we'd get a large positive y value. Thus as $x \to \infty$, $y \to \infty$

Resulting Shape

As $x \to \infty, y \to \infty$
As $x \to -\infty, y \to \infty$

$y = ax^3 + bx^2$
$+cx + d$

$y = ax^5 + bx^4 + \ldots$

(And we have the opposite if $a < 0$)

Resulting Shape

As $x \to \infty, y \to -\infty$
As $x \to -\infty, y \to -\infty$

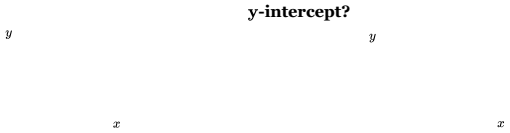
$y = ax^4 + bx^3$
$+cx^2 + dx + e$

If a>0, what therefore can we say about the shape if:

- The order is odd:
- The order is even:

Chapter 4 · Original slide 7 · Cubics

y-intercept?



Sketch the curve with equation
$y = x^2(x - 1)$

Shape?

Shape?

Roots?

y-intercept?

Sketch the curve with equation
$y = (x - 2)(1 - x)(1 + x)$

Features you must consider:

Tip: No need to expand out the whole thing. Just mentally consider the x terms multiplied together.

Roots?

This is sort of because the curve crosses at 0 then immediately crosses at 0 again!

Tip: It's incredibly easy to forget to write in one of the intercepts. So don't!

1.15 Graph Sketching: Cubic Graphs

Chapter 4 · Original slide 8 · Cubics



Sketch the curve with equation
$y = (2 - x)(x + 1)^2$

Sketch the curve with equation
$y = (x - 4)^3$

Shape?

Roots?

y-intercept?

Shape?

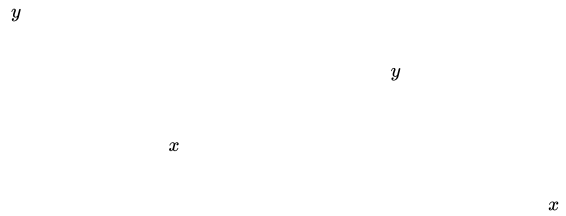
Roots?

y-intercept?

Exam Notes: The term ‘point of inflection’ has been removed from the new A Level syllabus (bad idea!!).

A point of inflection is where the curve goes from ‘convex’ to ‘concave’ (or vice versa), i.e. curves in one direction before and curves in another direction after. You might have encountered these terms in Physics.

Chapter 4 · Original slide 9 · Cubics with Limited Roots



Sketch the curve with equation
$y = (x + 1)\left(x^2 + x + 1\right)$

Shape?

y-intercept?

Roots?

However, in Chapter 12, we’ll be able to work turning points using ‘differentiation’, and hence conclude that it doesn’t have any!

1.15 Graph Sketching: Cubic Graphs

Chapter 4 · Original slide 10 · Finding the equation yourself

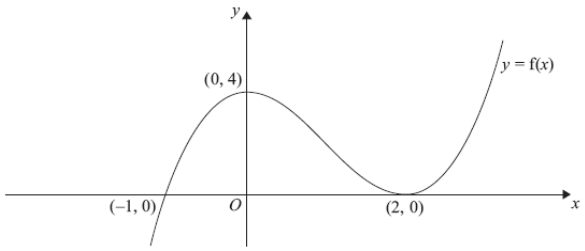


Figure 1 shows a sketch of the curve C with equation $y = f(x)$.

The curve C passes through the point $(-1, 0)$ and touches the x-axis at the point $(2, 0)$.

The curve C has a maximum at the point $(0, 4)$.

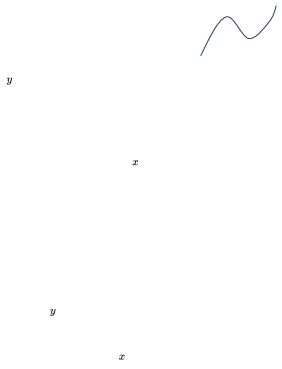
The equation of the curve C can be written in the form.

$y = x^3 + ax^2 + bx + c$

where a, b and c are integers.

(a) Calculate the values of a, b, c.

Chapter 4 · Original slide 11 · Test Your Understanding



1

Sketch the curve with equation

$y = x(x-3)^2$

2

Sketch the curve with equation

$y = -(x+2)^3$

3

A curve has this shape, touches the x axis at 3 and crosses the x axis at -2. Give a suitable equation for this graph.

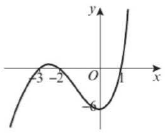
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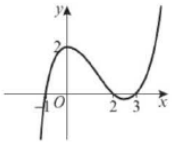
Sketch the curve with equation

$y = 2x^2(x-1)(x+1)^3$

1.15 Graph Sketching: Cubic Graphs

Chapter 4 · Original slide 12 · Exercise 4A –Y1 Pure pages 63





5 The graph of $y = x^3 + bx^2 + cx + d$ is shown opposite, where b,c,d are real constants.

a Find the values of b, c, d. (3 marks)
b Write down the coordinates of the point where the curve crosses the y-axis. (1 mark)

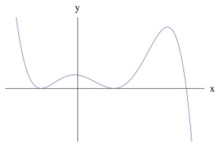
Chapter 4 · Original slide 13 · Exercise 4A –Y1 Pure pages 63

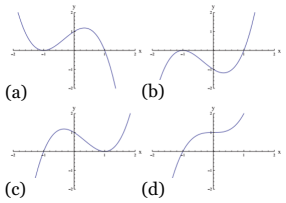
7 Given that $\mathrm{f}(x) = (x - 10)(x^2 - 2x) + 12x$

a Express $\mathrm{f}(x)$ in the form $x(ax^2 + bx + c)$ where a, b, c are real constants.
b Hence factorise $\mathrm{f}(x)$ completely.
c Sketch the graph of $y = \mathrm{f}(x)$ showing clearly the points where the graph intersects the axes.

1.15 Graph Sketching: Cubic Graphs

Chapter 4 · Original slide 14 · Exercise 4A Extension





Extension

2

[MAT 2011 1A] A sketch of the graph $y = x^3 - x^2 - x + 1$ appears on which of the following axis?

A) $y = (3-x)^2(3+x)^2(1-x)$

B) $y = -x^2(x-9)\left(x^2-3\right)$

C) $y = (x-6)(x-2)^2(x+2)^2$

D) $y = \left(x^2-1\right)^2(3-x)$

1

[MAT 2012 1E] Which one of the following equations could possibly have the graph given below?

Cubics can sometimes be factorised by pairing the terms:

Chapter 4 · Original slide 15 · Recap

a x

b x

c x

If we sketched $y = (x-a)(x-b)^2(x-c)^3$ what happens on the x-axis at:

$x = a$:

$x = b$:

$x = c$: